

Lab 31 — Saving and Loading Data with NumPy

Aim

To save NumPy arrays to disk in binary and text formats and reload them for later use.

Algorithm / Procedure

1. Import NumPy.
2. Create a NumPy array.
3. Save the array in binary format using `np.save()`.
4. Save the array in CSV format using `np.savetxt()`.
5. Load the binary file using `np.load()`.
6. Load the CSV file using `np.loadtxt()`.
7. Display the loaded arrays.

Program

```
import numpy as np

a=np.array([[1,2,3],[4,5,6]])

np.save('array.npy',a)

np.savetxt('array.csv',a,delimiter=',')

loaded_numpy=np.load('array.npy')

loaded_csv=np.loadtxt('array.csv',delimiter=',')

print("Loaded from .npy:\n",loaded_numpy)

print("Loaded from .csv:\n",loaded_csv)
```

Output

Loaded from .npy:

```
[[1 2 3]
```

```
[4 5 6]]
```

Loaded from .csv:

```
[[1. 2. 3.]
```

```
[4. 5. 6.]]
```

Result

NumPy arrays were successfully saved and loaded using both binary .npy and text .csv file formats.